

CPF Specification / 기술 명세

Public API·Annotation·Starter·Config·State 를 빠르게 찾는 기술 Reference

Core Platform Framework

기준: 최신 로컬 Source / 2026-08-16

전체 목차

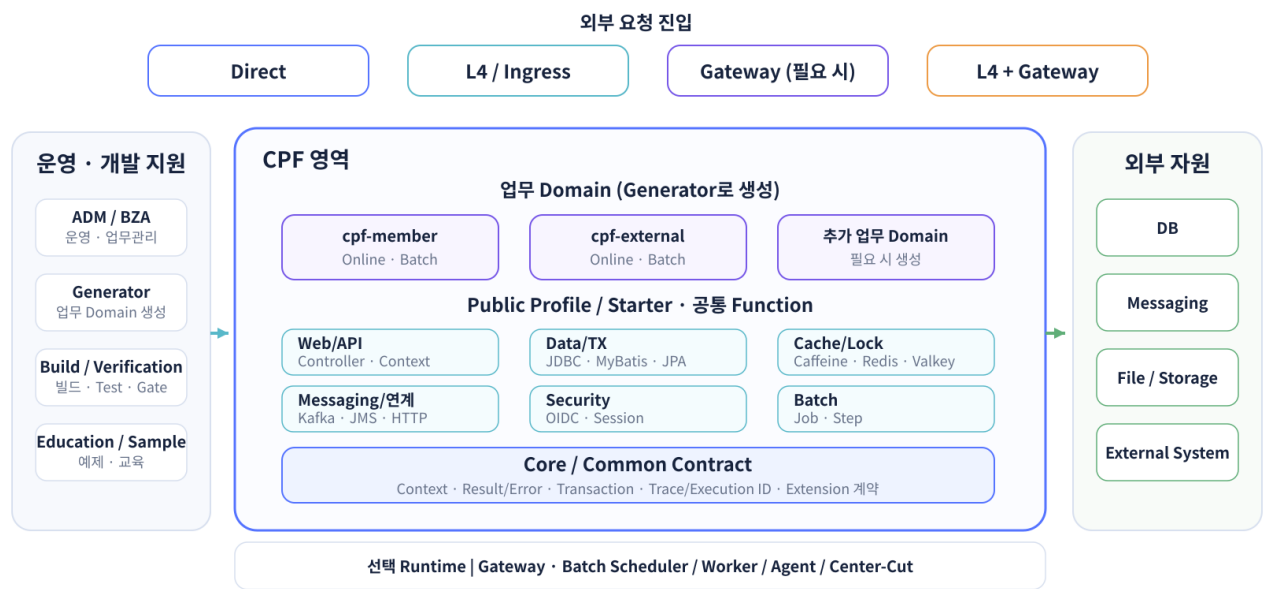
장	찾을 수 있는 내용
1. Architecture·Ownership	CPF 영역/Module 책임과 공개 경계
2. Public Profile·Starter	사용자가 선택할 Artifact/Provider
3. Web·Controller	@CpfController/CpfBaseController 계약
4. Service	@CpfService/CpfBaseService 계약
5. Persistence·Transaction	Repository/DAO/@CpfTx 계약
6. Cache·Lock	Cache 공개 API/옵션
7. Domain·Integration	CpfDomainClient/CpfResult/ServiceCall
8. Messaging	Publisher/Listener 계약
9. Security·Logging·Audit	Annotation 계약
10. Common Function	Code/Parameter/Message/Calendar API
11. Batch 공개 계약	Job/Step/Runtime 공개 경계
12. Gateway·Runtime	Gateway/ADM Runtime 경계
13. Command·Config·DB	개발 명령/환경/DB Vendor
14. Error·State·UNKNOWN	표준 결과/운영 상태
15. Source Index	정확한 Source 위치

1. Architecture·Ownership

[전체 목차로](#)

CPF 전체 Architecture

외부 진입부터 업무 Domain, Public Starter, 공통 계약, 선택 Runtime, 운영·개발 지원과 외부 자원 경계를 한눈에 봅니다.



영역	공식 책임	사용자 경계
cpf-core	Topology-independent 핵심 계약	Context/Result/Error/Domain Client 계약
cpf-starters	Public Profile/Starter/Provider/API	업무 Domain 이 선택/사용
cpf-admin	플랫폼 운영·관리	ADM
cpf-biz-admin	업무 관리자 기능	필요 시 사용하는 BZA
cpf-batch	Batch Runtime	Scheduler/Worker/Agent/Center-Cut
cpf-gateway	외부 진입 Runtime	필요 시 사용하는 Gateway
cpf-tools	Generator/Build/DB/Verification	개발/배포 도구

2. Public Profile·Starter

[전체 목차로](#)

분류	Public 선택	용도
Profile	cpf-starter-web-api	일반 REST API
Profile	cpf-starter-secure-api	인증/권한 API
Profile	cpf-starter-bff	Browser BFF
Profile	cpf-starter-event	이벤트 중심 서비스
Profile	cpf-starter-batch	Batch 구성
Data	cpf-starter-data-jdbc/mybatis/jpa	Persistence Provider
Cache	cpf-starter-cache-caffeine/redis/valkey	Local/Shared Cache
Messaging	cpf-starter-messaging-kafka/rabbitmq/jms/ibm-mq	Broker Provider
Session	cpf-starter-session-jdbc/valkey	Session Provider
기타	cpf-starter-object-storage-s3 / graphql / realtime / oidc	선택 기능

Internal Starter Catalog 의 internalOnlyArtifacts 는 Generator 로 생성한 업무 Domain 이 직접 의존하는 Public 선택 Surface 가 아닙니다.

3. Web·Controller

[전체 목차로](#)

API	계약	주요 멤버/옵션
@CpfController	CPF Web Controller 등록 + RestController	contextRequired=true
CpfBaseController	Context/Validation/Response helper	requireCurrentContext, requireContext, requireText, requireValue, requireRule, ok, created, noContent, executionFacts

4. Service

[전체 목차로](#)

API	계약	주요 멤버/옵션
@CpfService	CPF Business Service 등록	contextRequired=true
CpfBaseService	최소 공통 Service Base	requireText
Domain Base Service	Generator Domain 공통 Helper	requireTransactionId, transactionSequence, actorId 등 Domain 별 생성 내용

5. Persistence·Transaction

[전체 목차로](#)

API	계약	주요 멤버/옵션
@CpfRepository	Persistence Repository 등록	contextRequired=true
@CpfDao	JDBC/MyBatis Class DAO 등록	contextRequired=true
CpfBaseRepository	Context/Validation/Paging helper	statementId, boundedSize, executionFacts
CpfBaseDao	DAO Base helper	statementId, boundedSize, executionFacts
@CpfTx	CPF Transaction 선언	id,name,ownerDomain,transactionManager,propagation,isolation,readOnly,timeoutSeconds 등

6. Cache·Lock

[전체 목차로](#)

API	Method/옵션	계약
CpfCachePort	get/put/evict/evictNamespace/metrics/health	Provider-neutral Cache Port
CpfCacheAsideService	getOrLoad(key,options,loader)	Cache-aside + single-flight
CpfCacheOptions	ttl, negativeTtl, lockWait, lockLease, cacheNull, failOpen	만료/Negative/장애 정책
CpfDistributedLockPort	tryAcquire/release	분산 Lock 경계

7. Domain·Integration

[전체 목차로](#)

API	Method/상태	계약
CpfDomainClient<I,O>	execute(I) → CpfResult<O>	Topology-independent Typed Domain Client
CpfResult<T>	SUCCESS/BUSINESS_FAILURE/TECHNICAL_FAILURE/UNKNOWN	분산 Boundary 표준 Result
CpfServiceCaller	invoke(CpfServiceRequest, transport)	Registry/Health/Retry/Failover/Circuit/UNKNOWN 공개 진입점
CpfServiceRequest	serviceId, endpointCode, instanceId, method, path, timeoutMillis, retryCount, headers, attributes	외부/서비스 호출 요청
CpfServiceResult<T>	success/businessFailure/technicalFailure/unknown/recoveryRequired	서비스 호출 공개 결과

8. Messaging

[전체 목차로](#)

API	Method/옵션	계약
CmnMessagePublisher	publish(key,payload) / publish(destination,key,payload,headers)	Broker-neutral 업무 발행
@CpfMessageListener	destination, consumerGroup, idempotencyRequired, contextRequired	Provider 종속 Listener 와 분리된 정책
CpfBrokerBridgePort	publish/subscribe/findRecent	Provider adapter 경계

9. Security·Logging·Audit

[전체 목차로](#)

Annotation	주요 옵션	계약
@CpfPermission	value[], all=true	Authority fail-closed 검사
@CpfApprovalRequired	action, approvals, reasonRequired, approvalId/reason parameter	위험 동작 승인 요구
@CpfLogging	operation, mode, includeArguments/allowlist, includeResult/resultAllowlist	Context correlation + masking 로그
@CpfAudit	action, reasonRequired, includeSafeResultSummary	변경/위험 작업 Audit

10. Common Function

[전체 목차로](#)

API	주요 메소드	계약
CpfCodeService	values/find/required	공통코드 조회
CpfParameterService	find/requiredValue/findValue/ requiredValue(key,type)	Typed 공통 파라미터
CpfMessageService	resolve(code,locale,args)	Locale/Argument 메시지
CpfCalendarService	isBusinessDay/shiftBusinessDay/next/previous	영업일 API

11. Batch 공개 계약

[전체 목차로](#)

영역	공개 계약	주요 계약/용도
Annotation Job	com.cpf.batch.api.annotation.CpfBatchJob	value / restartable / maxConcurrentExecutions
Job Bean Metadata	com.cpf.batch.api.CpfBatchJob	id / name / ownerDomain; Spring Batch Job Bean 기반 구성
Step	com.cpf.batch.api.annotation.CpfBatchStep	value / order / idempotent
실행 요청	CpfBatchExecutionRequest	run / scheduledRun / retry / restart / rerun / stop / onDemand
Control Plane	BatchExecutionControlPort	start / stop / restart / abandon / reconcile
Topology	BatchExecutionTopology	LOCAL / PARALLEL_STEPS / LOCAL_PARTITION / REMOTE_PARTITION / REMOTE_CHUNK / REMOTE_STEP
상태	BatchControlState	RESERVED~COMPLETED / FAILED / UNKNOWN_RESULT / ABANDONED / REJECTED
실행 설정	cpf.batch.execution.*	Chunk/Partition/Remote/Executor 기본값·범위는 배치 개발자 가이드 §15

동일 **Annotation 명칭** `CpfBatchJob`은 package 에 따라 계약이 다릅니다. Annotation Job/Step 모델과 Job Bean metadata 모델의 import 와 속성을 구분합니다.

12. Gateway·Runtime

[전체 목차로](#)

영역	계약	참고
Gateway	Route/Target/Policy/Health/Connection Test	외부 진입 선택 Runtime
ADM	운영 조회/제어 API	권한·승인·Audit 포함
BZA	업무 관리자 기능	필요 시 사용
Local Runtime	cpfRunLocal	개발 통합 1 JVM/1 Port

13. Command·Config·DB

[전체 목차로](#)

분류	계약	값/명령
개발 Shell	cpf-tools\build\tools\cpf-dev.ps1	build/test/verify-fast/verify-full/run-local/run-batch/status/stop/modules/resource

분류	계약	값/명령
Root Gradle	cpf-root-conventions.gradle	cpfBuild/cpfTest/cpfVerifyFast/cpfVerifyFullLocal/ cpfRunLocal/Adm/Bza/Gateway/Batch/Education/ cpfModules
Resource Profile	gradle\cpf-runtime	common/local/dev/test/stg/prod.properties
DB Vendor	공식 Vendor	Oracle / PostgreSQL / MariaDB

14. Error·State·UNKNOWN

[전체 목차로](#)

상태	의미	사용 경계
SUCCESS	성공 확정	Domain/ServiceCall/Batch
BUSINESS_FAILURE	업무 실패 확정	업무 오류
TECHNICAL_FAILURE	기술 실패 확정	Runtime/Dependency
UNKNOWN	성공/실패를 확정할 수 없음	분산/외부 Boundary → Reconcile
PARTIAL	일부 Source 만 성공/가용	운영 통합 조회

15. Source Index

[전체 목차로](#)

계약	Source
CpfBaseController / @CpfController	cpf-starters/web/src/main/java/com/cpf/web/api
CpfBaseService / @CpfService	cpf-starters/base/runtime/src/main/java/com/cpf/foundation
Repository / @CpfTx	cpf-starters/data/persistence/src/main/java/com/cpf/data/persistence/api
Cache	cpf-starters/data/src/main/java/com/cpf/data/cache
Domain Result/Client	cpf-core/src/main/java/com/cpf/core/api
ServiceCall	cpf-starters/integration/src/main/java/com/cpf/integration/api/servicecall
Messaging	cpf-starters/messaging/src/main/java/com/cpf/messaging
Common	cpf-starters/common/src/main/java/com/cpf/common
Security	cpf-starters/security/src/main/java/com/cpf/security/api
Batch API	cpf-batch/api/src/main/java/com/cpf/batch/api
Batch Runtime	cpf-batch/runtime/src/main/java/com/cpf/batch/execution
Starter Catalog	cpf-tools/generator/contracts/cpf-starter-catalog.json